

VoxShield vs. iWAX – Architectural Comparison and Uniqueness

This document summarizes the key similarities and differences between the proposed VoxShield model and the iWAX (interpretable Wav2vec–AASIST–XGBoost) framework for voice spoofing detection. The goal is to clearly show the novelty of VoxShield for use in a research paper or project report.

1. Side-by-Side Architectural Comparison

Aspect	iWAX Framework	VoxShield (Proposed)
Core idea	Wav2vec2 front-end + AASIST anti-spoofing backbone + XGBoost classifier + event interpretability	Hybrid backbone with CNN + Transformer + attention pooling, multi-view fusion, and calibrated outputs (FCC/L)
Feature views	Primarily uses wav2vec2 embeddings produced by the AASIST-style front-end	Especially by multi-view fusion of SSL embeddings and spectrogram-based CNN features
Fusion mechanism	Pipeline-style: features flow from wav2vec2 early to AASIST, then to XGBoost	Early fusion: wav2vec2 and spectrogram features followed by Trans
Interpretability	Uses XGBoost feature importance to analyze influential regions in the embedding space	Uses attention weights in the transformer to highlight s
Training style	Trains wav2vec2–AASIST as a countermeasure backbone, XGBoost, and topi	Trains CNN backbone, XGBoost, and topi
Target scenario	Benchmark-oriented evaluation on ASVs	Focus on practical deepfake voice misuse (e.g., messaging
Deployment view	Primarily presented as a research countermeasure	Designed to be integrated into a web-based forensic dashbo

2. Claimed Uniqueness of VoxShield (for Research Paper)

- Multi-view fusion: VoxShield jointly learns from two distinct views of the audio (self-supervised wav2vec2 features and spectrogram-based CNN features) using early feature fusion and a shared attention backbone, instead of treating the anti-spoofing countermeasure and classifier as separate stages.
- Forensic-oriented outputs: VoxShield is designed to provide calibrated real/fake probabilities, discrete risk bands (e.g., low/medium/high), and time-localized suspicion maps, making it easier for human analysts to interpret model decisions.
- Practical deployment focus: While iWAX is evaluated primarily as a benchmark countermeasure, VoxShield targets real-world misuse scenarios such as deepfake voice messages and phone calls, and is intended to be embedded in a web-based dashboard.
- Modular backbone: VoxShield uses a modular CNN + Transformer + attention pooling design, making it easier to adapt, compress, or extend to new datasets and languages compared to tightly coupled AASIST + XGBoost pipelines.

These points can be directly used in the "Contributions" or "Novelty" section of your project report or research paper to clearly distinguish VoxShield from iWAX.